SARS-CoV-2 VARIANT IDENTIFICATION IN ACCRA METROPOLIS POST-LOCKDOWN: PREVALENCE, ASSOCIATIONS AND IMPLICATIONS

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Introduction

- ❑ COVID-19 caused by the SARS-CoV-2 was declared a global pandemic by the WHO in 2020.
- ❑ Post-lockdown adherence to preventive measures has significantly reduced leading to the resurgence of the virus.
- ❑ Surveillance of high-risk areas requires the detection of infection and circulating variants within the area.
- ❑ Currently, sequencing is the standard for variant identification and it is known to be expensive, laborious, and requires high technical skills
- ❑ This study leverages PCR techniques for the identification of SARS-CoV-2 variants within the Accra Metropolis post-lockdown.

Objectives

- ❑ To determine the prevalence and severity of SARS-CoV-2 infection within Accra Metropolis post-lockdown.
- ❑ To identify circulating variants within the Metropolis.
- ❑ To evaluate the risk factors and predictors associated with SARS-CoV-2 post-lockdown.

Methodology

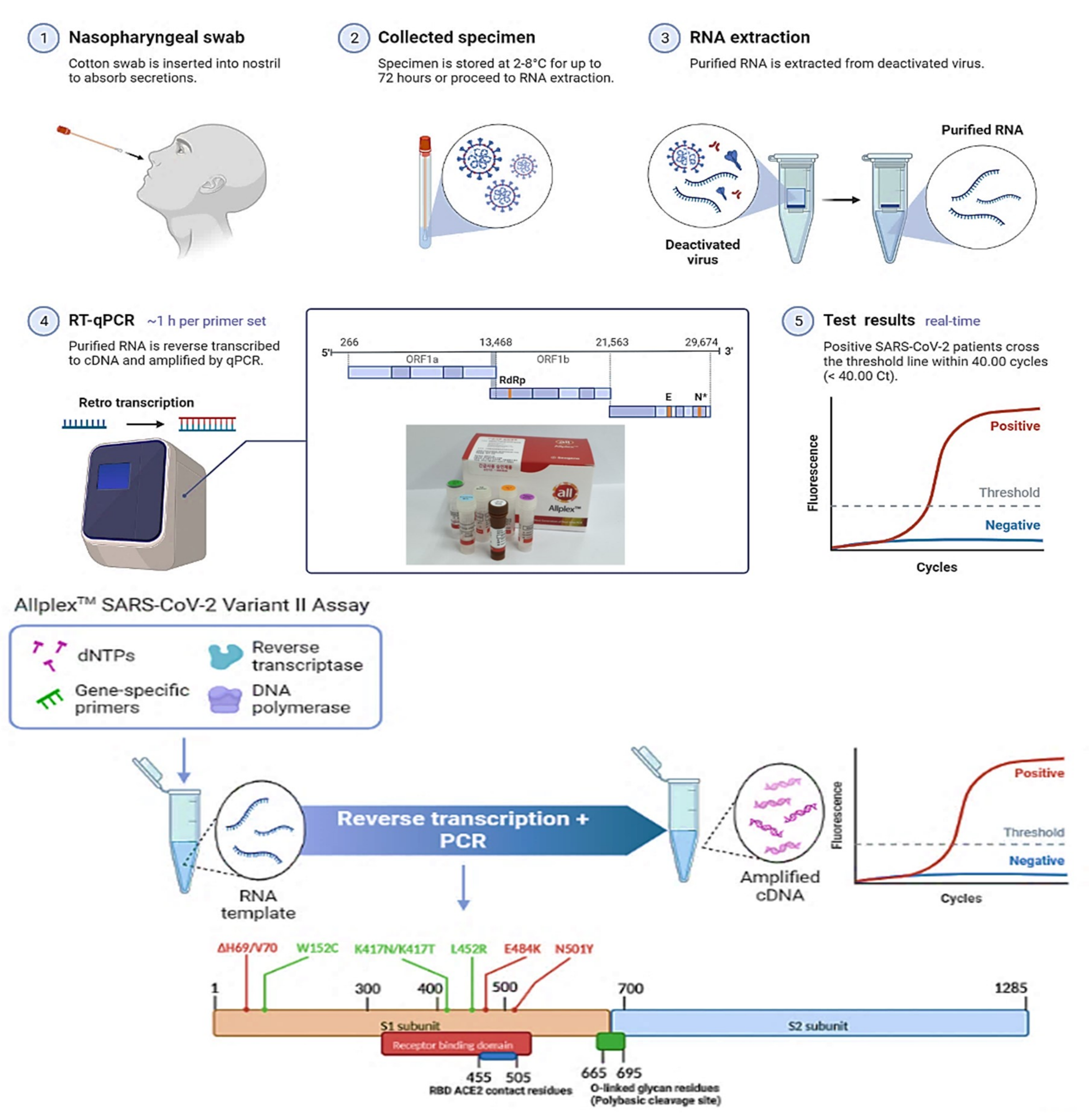

Fig. 1: Sample Collection and Laboratory Analysis

Results

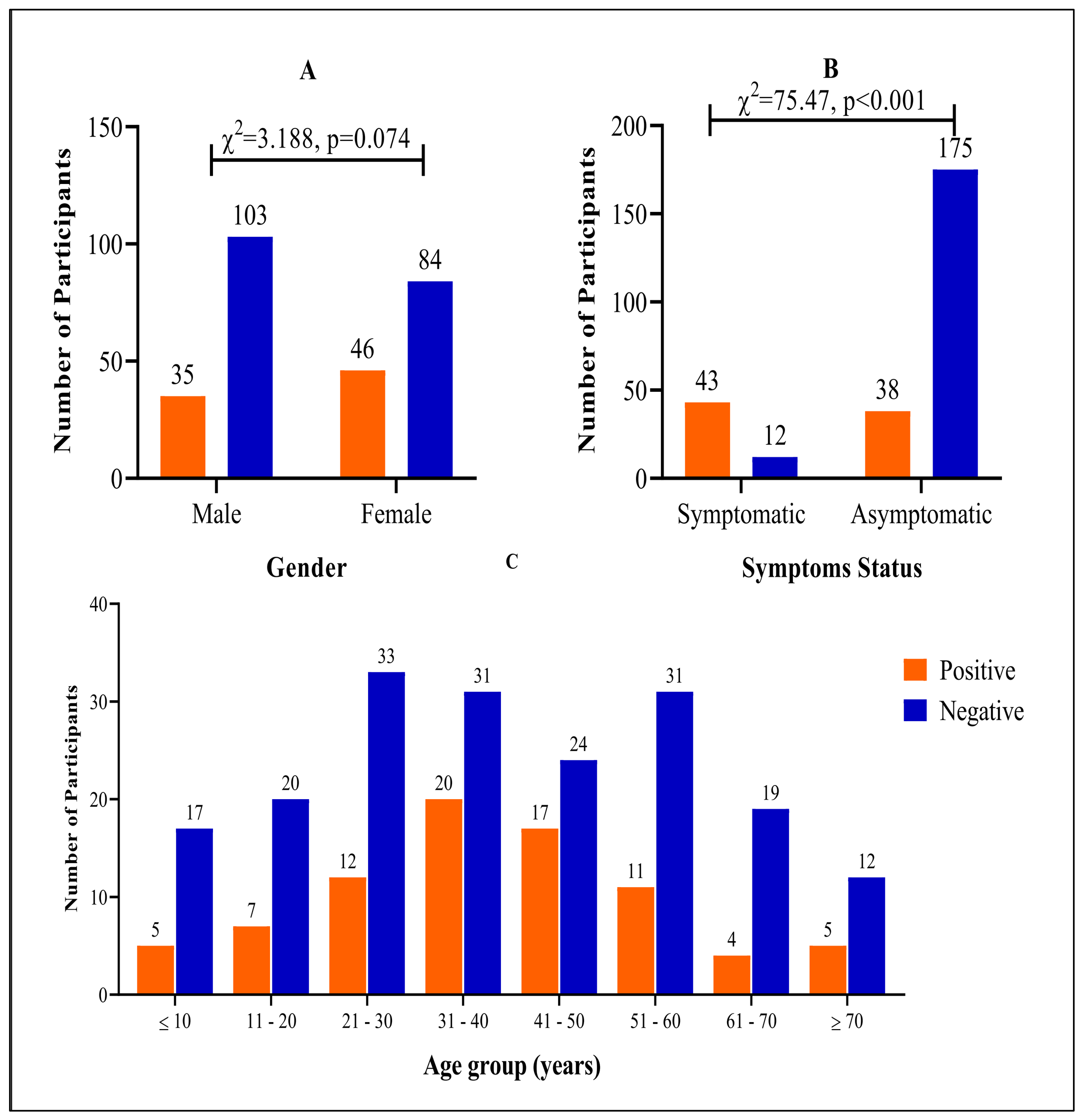

Fig. 2: Distribution of SARS-CoV-2 infection (A) Gender, (B) Symptoms, (C) Age groups

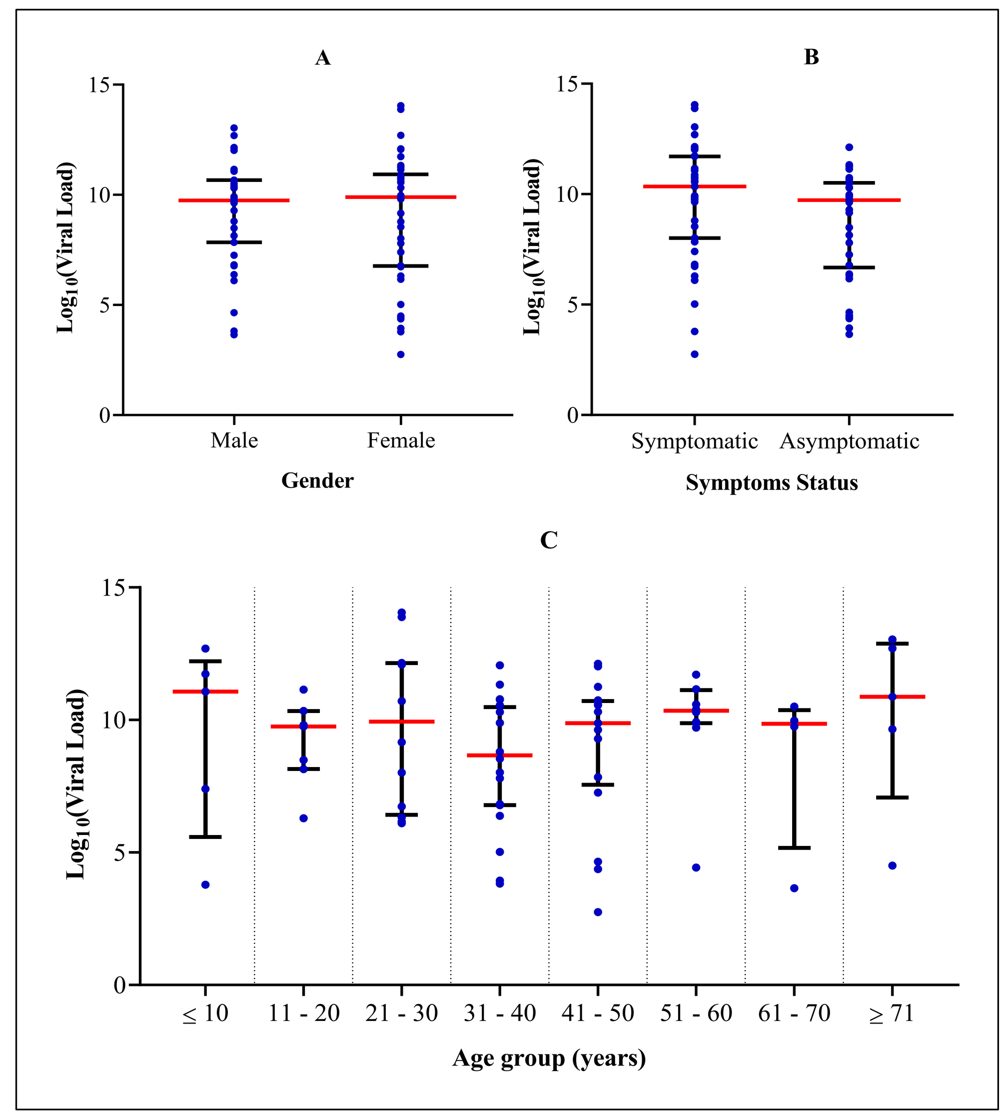

Fig. 3: Distribution of SARS-CoV-2 viral load (A) Gender (B) Symptoms, (C) Age group

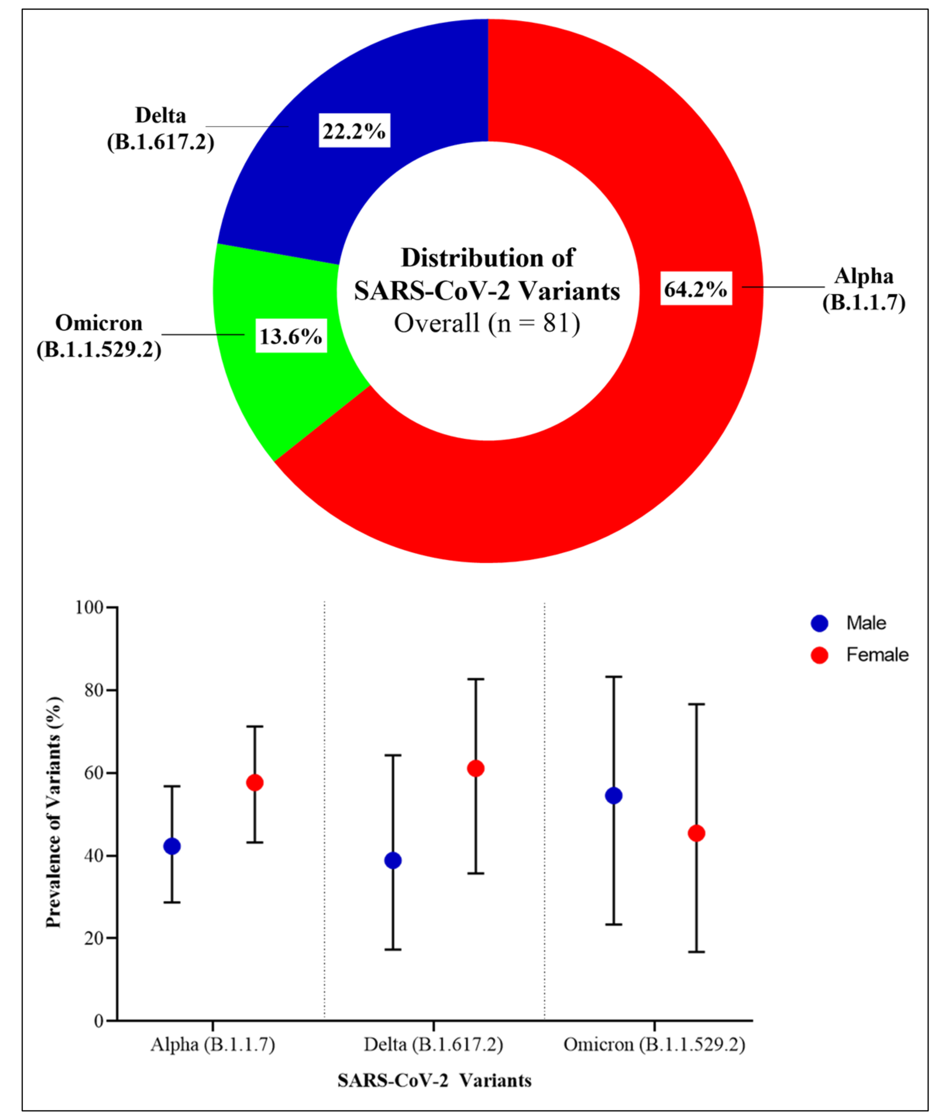

Fig. 4: Circulating SARS-CoV-2 Variants

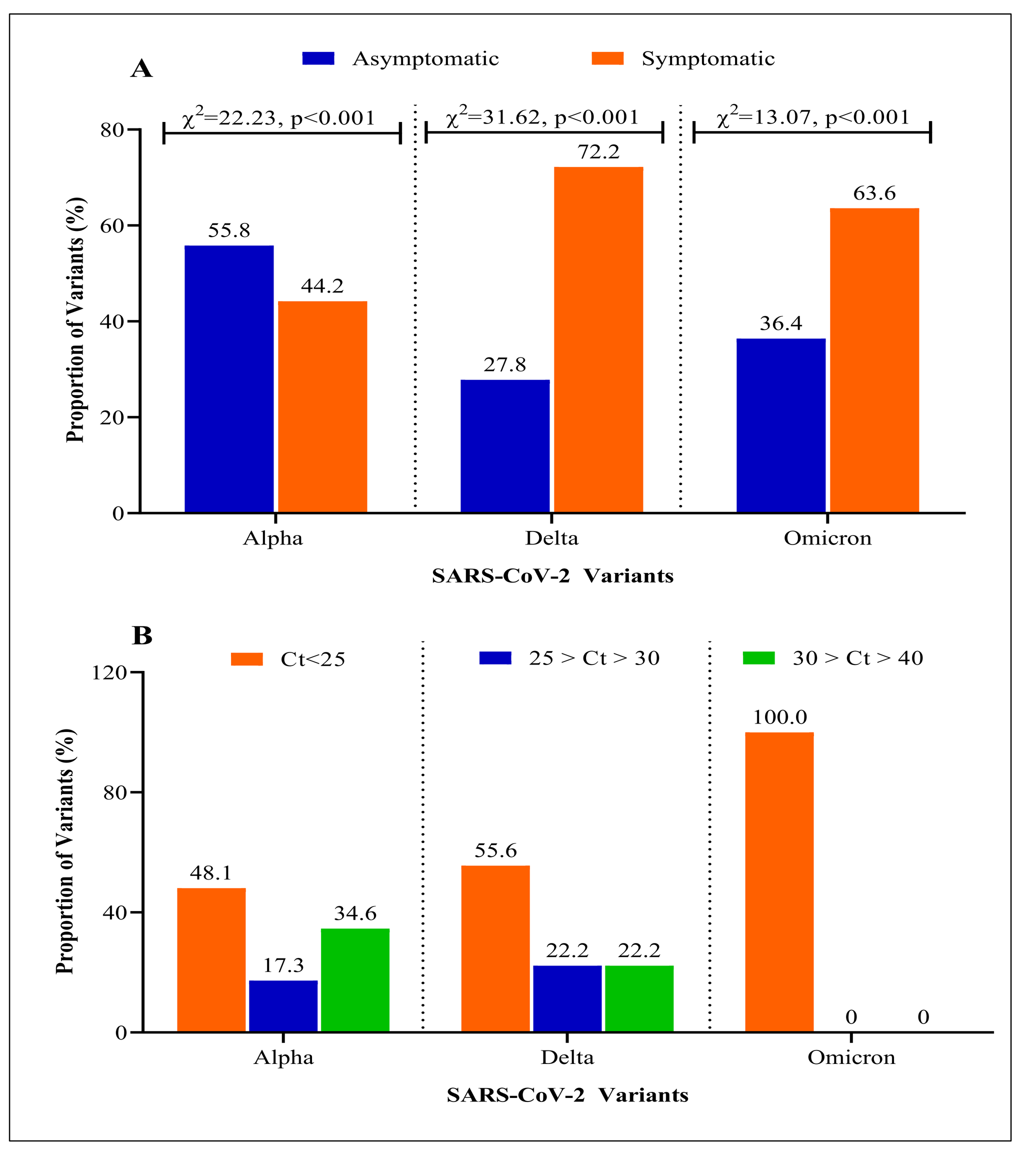

Fig. 5 Distribution of SARS-CoV-2 variant (A) Symptom Status, (B) Viral Load

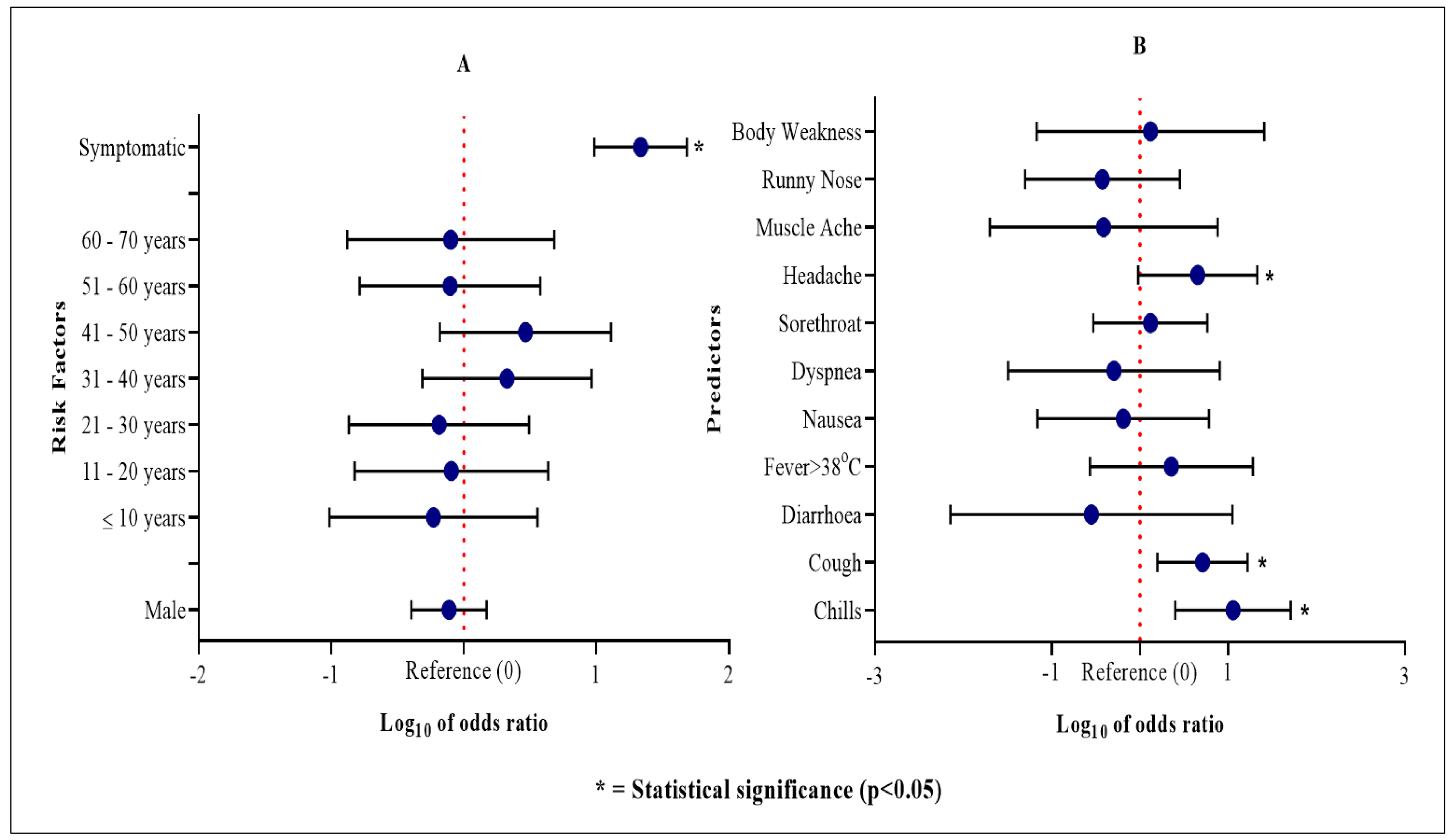

Fig. 6: Risk factors (A) and Predictors (B) associated with SARS-CoV-2 Infection

Conclusion

- ❑ There was an increase in SARS-CoV-2 cases within the Accra Metropolis post-lockdown compared to previous studies.
- ❑ High prevalence of SARS-CoV-2 infection within the age group of 31 - 40 years, but participants aged ≤10 years have the highest viral load.
- ❑ The alpha variant was the predominant circulating variant and the omicron variant was associated with high viral load during the study period.
- ❑ Severe symptoms of SARS-CoV-2 were associated with the Omicron variant infections.
- ❑ Chills and cough were symptoms strongly associated with SARS-CoV-2 infection.

Acknowledgements

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